EXPERIMENT-2

Aim:- Convert the ER Diagram into a relational schema. Write complete MySQL CREATE TABLE statements with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and SET NULL constraints. Insert sample data and demonstrate referential integrity violations

Source Code :-

1. Relational Schema

Table	Primary Key	Important Foreign Keys
CUSTOMER	CustomerID	—
CUSTOMER_PHONE	(CustomerID, Phone)	CustomerID → CUSTOMER
SELLER	SellerID	—
SELLER_PHONE	(SellerID, Phone)	SellerID → SELLER
ADDRESS	AddressID	CustomerID → CUSTOMER
CATEGORY	CategoryID	PARENTCATEGORYID → CATEGORY
PRODUCT	ProductID	SellerID → SELLER, CategoryID-CATEGORY
ELECTRONICS	ProductID	—
CLOTHING	ProductID	—
BOOKS	ProductID	—
ORDERS	OrderID	CustomerID-XCUSTOMER
ORDERITEM	(OrderID, ItemNo)	OrderID → ORDERS
PAYMENT	PaymentID	OrderID → ORDERS
DELIVERY	DeliveryID	OrderID → ORDERS

2. SQL Queries

```
CREATE DATABASE ECOMMERCE;
USE ECOMMERCE;

CREATE TABLE CUSTOMER(
    CustomerID INT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    Email VARCHAR(100) NOT NULL UNIQUE,
    Password VARCHAR(100) NOT NULL,
    DateOfBirth DATE NOT NULL);

CREATE TABLE CUSTOMER_PHONE(
    CustomerID INT,
    Phone VARCHAR(15),
    PRIMARY KEY (CustomerID, Phone),
    FOREIGN KEY (CustomerID) REFERENCES CUSTOMER,
    (CustomerID) ON DELETE CASCADE);

CREATE TABLE SELLER(
    SellerID INT PRIMARY KEY,
    SellerName VARCHAR(50) NOT NULL,
    Email VARCHAR(100) UNIQUE NOT NULL,
    GSTIN VARCHAR(20) UNIQUE NOT NULL,
    AccountNo VARCHAR(20),
    IFSCCode VARCHAR(15),
    BankName VARCHAR(50));
```

```
CREATE TABLE SELLER_PHONE (
    SellerID INT,
    Phone VARCHAR(15),
    PRIMARY KEY (SellerID, Phone),
    FOREIGN KEY(SellerID) REFERENCES SELLER
    (SellerID) ON DELETE CASCADE);

CREATE TABLE CATEGORY (
    CategoryID INT PRIMARY KEY,
    CategoryName VARCHAR(50) NOT NULL,
    Description VARCHAR(100),
    ParentCategoryID INT,
    FOREIGN KEY (ParentCategoryID) REFERENCES
    CATEGORY (CategoryID) ON DELETE SET NULL);

CREATE TABLE PRODUCT (
    ProductID INT PRIMARY KEY,
    ProductName VARCHAR(50) NOT NULL,
    Description VARCHAR(500),
    Price DECIMAL(10,2) NOT NULL,
    StockQty INT NOT NULL,
    Weight DECIMAL(8,2)
    SellerID INT NOT NULL, CategoryID INT,
    FOREIGN KEY (SellerID) REFERENCES SELLER
    (SellerID) ON DELETE CASCADE,
    FOREIGN KEY (CategoryID) REFERENCES CATEGORY
    (CategoryID) ON DELETE CASCADE);
```

```
CREATE TABLE ADDRESS(
        AddressID INT PRIMARY KEY,
        DoorNo VARCHAR(30),
        Street VARCHAR(50),
        Area VARCHAR(50),
        City VARCHAR(30),
        State VARCHAR(30),
        Pincode VARCHAR(10) NOT NULL,
        AddressType VARCHAR(20),
        CustomerID INT NOT NULL,
        FOREIGN KEY(CustomerID) REFERENCES CUSTOMER
        (CustomerID) ON DELETE CASCADE);

CREATE TABLE ORDERS(
        OrderID
        OrderDate
        OrderStatus
        TotalAmount
        CustomerID
        AddressID
        FOREIGN KEY(CustomerID) REFERENCES CUSTOMER
        (CustomerID) ON DELETE CASCADE,
        FOREIGN KEY (AddressID) REFERENCES ADDRESS
        (AddressID) ON DELETE SET NULL);
```

```
CREATE TABLE ELECTRONICS (
    ProductID INT PRIMARY KEY,
    Brand VARCHAR(50),
    Warranty VARCHAR(50),
    PowerRating VARCHAR(50),
    FOREIGN KEY(ProductID) REFERENCES PRODUCT
    (ProductID) ON DELETE CASCADE );

CREATE TABLE CLOTHING (
    ProductID INT PRIMARY KEY,
    Size VARCHAR(20),
    Colour VARCHAR(20),
    Material VARCHAR(30),
    FOREIGN KEY(ProductID) REFERENCES PRODUCT
    (ProductID) ON DELETE CASCADE);

CREATE TABLE BOOKS (
    ProductID
    ISBN VARCHAR(20) UNIQUE,
    Author VARCHAR(50),
    Publisher VARCHAR(50),
    FOREIGN KEY(ProductID) REFERENCES PRODUC
    (PRODUCTID) ON DELETE CASCADE);
```

```
CREATE TABLE ORDERITEM (
    OrderID INT,
    ItemNo INT,
    ProductID INT NOT NULL,
    ItemPrice DECIMAL(10,2) NOT NULL,
    Quantity INT NOT NULL,
    Discount DECIMAL(10,2),
    TaxAmount DECIMAL(10,2),
    PRIMARY KEY (OrderID, ItemNo),
    FOREIGN KEY (OrderID) REFERENCES ORDERS
    (OrderID) ON DELETE CASCADE,
    FOREIGN KEY (ProductID) REFERENCES PRODUCT
    (ProductID) ON DELETE CASCADE);

CREATE TABLE PAYMENT(
    PaymentID INT PRIMARY KEY,
    PaymentDate DATE NOT NULL,
    Amount DECIMAL(10,2) NOT NULL,
    PaymentMethod VARCHAR(30) NOT NULL,
    TransactionID VARCHAR(30) UNIQUE,
    PaymentStatus VARCHAR(20) NOT NULL,
    OrderID INT UNIQUE NOT NULL,
    FOREIGN KEY (OrderID) REFERENCES ORDERS
    (OrderID) ON DELETE CASCADE);
```

```
CREATE TABLE DELIVERY (
    DeliveryID INT PRIMARY KEY,
    DeliveryDate DATE,
    DeliveryStatus VARCHAR(20) NOT NULL,
    TrackingID VARCHAR(30) UNIQUE,
    DeliveryPartner VARCHAR(50),
    ShippingCharge DECIMAL(8,2),
    OrderID INT UNIQUE NOT NULL,
    AddressID INT,
    FOREIGN KEY(OrderID) REFERENCES ORDERS
    (OrderID) ON DELETE CASCADE,
    FOREIGN KEY (AddressID) REFERENCES ADDRESS
    (AddressID) ON DELETE SET NULL);

INSERT INTO CUSTOMER VALUES
(1, 'Umati', 'umati10mishra@gmail.com', 'pass123', '2006-01-10');

INSERT INTO CUSTOMER_PHONE VALUES
(1, '9214293443');

INSERT INTO SELLER VALUES
(101, 'TechStore', 'tech@gmail.com', 'GST123', '123456', 'SBIN001', 'SBI
);

INSERT INTO SELLER_PHONE VALUES
(101, '9213413345');
```

INSERT INTO CATEGORY VALUES
(201, 'Electronics', 'Electronic items', NULL);

INSERT INTO CATEGORY VALUES
(202, 'Mobiles', 'Mobile Phones', 201);

INSERT INTO PRODUCT VALUES
(301, 'Laptop', 'Gaming Laptop', 75000, 10, 2.5, 101, 201);

INSERT INTO ELECTRONICS VALUES
(301, 'ASUS', '2 Years', '180W');

INSERT INTO ADDRESS VALUES
(401, '12', 'Eldeo', 'Transport Nagar', 'Lucknow',
'Uttar Pradesh', '160017', 'Home', 1);

INSERT INTO ORDERS VALUES
(501, '2026-08-14', 'Placed', 75000, 1, 401);

INSERT INTO ORDERITEM VALUES
(501, 1, 301, 75000, 1, 1000, 9000);

INSERT INTO DELIVERY VALUES
(701, NULL, 'Processing', 'TRK123',
'Delhivery', 100, 501, 401);

INSERT INTO PAYMENT VALUES
(601, 2026-08-14, 74000, 'UPI', 'TXN123', 'Paid', 501);

--REFERENTIAL INTEGRITY VIOLATION

INSERT INTO ORDERS VALUES

(502, '2026-08-14', 'Placed', 50000, 999, 401

Output :-

Output 1 - SHOW TABLES

Query -

```
SHOW TABLES;
```

Result -

Tables_in_ecommerce
ADDRESS
BOOKS
CATEGORY
CLOTHING
CUSTOMER
CUSTOMER_PHONE
DELIVERY
ELECTRONICS
ORDERITEM
ORDERS
PAYMENT
PRODUCT
SELLER
SELLER_PHONE

Output 2 - Sample Data

Query -

```
SELECT * FROM CUSTOMER;
SELECT * FROM ORDERS;
```

Result -

Customer

CustomerID	Name	Email	Password	DateOfBirth
1	Unnati	unnatiomishra@gmail.com	pass123	2006-01-10

Orders

OrderID	OrderDate	OrderStatus	TotalAmount	CustomerID	AddressID
501	2026-08-14	Placed	75000.00	1	401

Output 3 - Referential Integrity Violation

Query -

```
INSERT INTO ORDER VALUES
(502, '2026-08-14', 'Placed', 5000, 999, 401);
```

Result -

ERROR 1452 (23000):
Cannot add or update a child row:
a foreign key constraint fails

Reason - CustomerID 999 does not exist in CUSTOMER

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